Expt. No. _____ Page No. _____

Tutorial - 10

(1) (a) $\int_0^\infty |R(r)|^2 r^2 \, dr = 1.$

$\Rightarrow |A|^2 \int_0^\infty r^2 e^{-2r/a_0} \, dr = 1.$

~~$\Rightarrow$ $|A|^2$~~ Let $\frac{2r}{a_0} = t \Rightarrow dr = \frac{a_0}{2} dt$

$\therefore \int_0^\infty r^2 e^{-2r/a_0} \, dr = \int_0^\infty \frac{a_0^2}{4} t^2 \cdot e^{-t} \, \cancel{dt} \cdot \frac{a_0}{2} dt.$

$= \frac{a_0^3}{8} \Gamma(3) = \frac{a_0^3}{4}$

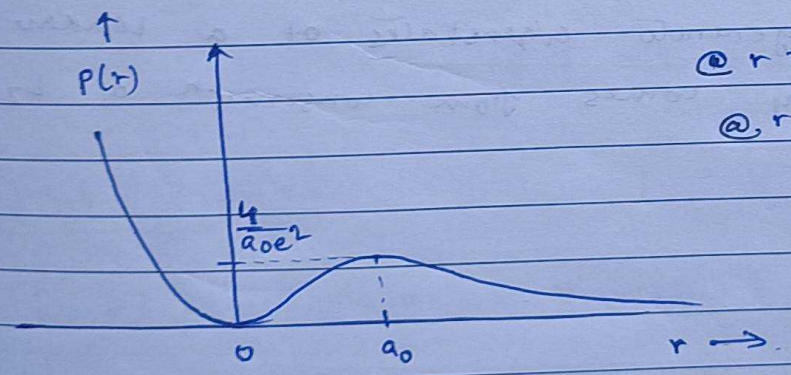
$\therefore |A| = \frac{2}{} \left(\frac{4}{a_0^3}\right)^{1/2} = \frac{2}{a_0^{3/2}} = \boxed{2a_0^{-3/2} = |A|}$ (Ans)

(b) $P(r) = |R|^2 r^2$

$= \frac{4}{a_0^3} e^{-2r/a_0} \cdot r^2$

$\therefore \frac{dP(r)}{dr} \Rightarrow e^{-2r/a_0} \left[2r - \frac{2r^2}{a_0} \right] = 0.$

$\Rightarrow r = \frac{r^2}{a_0} \Rightarrow \boxed{\cancel{r = a_0}}$ $\boxed{r = 0, a_0}$



@ $r = 0 \rightarrow P(0) = 0. \rightarrow$ minima

@, $r = a_0 \rightarrow P(a_0) = \frac{4}{a_0} e^{-2}$ (maxima)

(c) $r_{mp} = a_0$ as derived above.

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② $\int_0^{a_0} |R|^2 r^2 \, dr = \int_0^{a_0} \frac{4}{a_0^3} \cdot e^{-2r/a_0} \, r^2 \, dr$

$= \frac{4}{a_0^3} \int_0^{a_0} e^{-2r/a_0} \, r^2 \, dr.$

Let $t = \frac{2r}{a_0} \Rightarrow dt \cdot \frac{a_0}{2} = dr.$

$\therefore \int_0^{a_0} |R|^2 r^2 \, dr = \frac{4}{a_0^3} \int_0^2 e^{-t} \, t^2 \frac{a_0^2}{4} \cdot dt \cdot \frac{a_0}{2}$

$= \frac{1}{2} \int_0^2 e^{-t} \cdot t^2 \, dt$

$= \frac{1}{2} \left[-t^2 e^{-t} \Big|_0^2 + 2 \int_0^2 t \, e^{-t} \cdot dt \right]$

$= \frac{1}{2} \left[-4e^{-2} + 2 \left[-t e^{-t} \Big|_0^2 + \int_0^2 e^{-t} \, dt \right] \right]$

$= \frac{1}{2} \left[-4e^{-2} - 4e^{-2} \cancel{+ 2e^{-2} - 2} \right]$
$\cancel{+ 2 e^{-t} \Big|_0^2}$

$= \frac{1}{2} \left[-8e^{-2} - 2e^{-2} + 2 \right] = 1 - 5e^{-2}$

$\therefore \boxed{P_r(r < a_0) = 1 - 5e^{-2}}$ (Ans)

③ $\cancel{\langle r \rangle} \int_0^\infty r^n e^{-kr} \, dr.$

Let $kr = t \Rightarrow dt = k\,dr \Rightarrow dr = \frac{dt}{k}$

$\therefore \int_0^\infty \frac{t^n}{k^n} \cdot e^{-t} \frac{dt}{k} = \frac{1}{k^{n+1}} \int_0^\infty t^n e^{-t} \, dt$

$$\boxed{\int_0^\infty r^n e^{-kr} \, dr = \frac{n!}{k^{n+1}}}$$

$k = \frac{2}{a_0}.$

$\therefore \langle r \rangle = \int_0^\infty r \, P(r) \, dr = \frac{4}{a_0^3} \int_0^\infty r^3 e^{-2r/a_0} \, dr$

$\boxed{\langle r \rangle = \frac{3}{2} a_0}$

$= \frac{4}{a_0^3} \cdot \frac{3!}{\left(\frac{2}{a_0}\right)^4} = \frac{\cancel{4} \cdot 6}{\cancel{16}\, 4} a_0 = \frac{3}{2} a_0.$

Expt. No. _____ Page No. _____

$$\therefore \left\langle \frac{1}{r} \right\rangle = \int_0^\infty \frac{1}{r} P(r)\, dr$$

$$= \frac{4}{a_0^3} \int_0^\infty r\, e^{-2r/a_0}\, dr$$

$$= \frac{4}{a_0^3} \frac{1}{\left(\frac{2}{a_0}\right)^2} = \frac{1}{a_0} \quad \text{(Ans)}$$

$$\boxed{\left\langle \frac{1}{r} \right\rangle = \frac{1}{a_0}} \quad \text{(Ans)}$$

$$\therefore \langle r^2 \rangle = \int_0^\infty r^2 P(r)\, dr$$

$$= \frac{4}{a_0^3} \int_0^\infty r^4 e^{-2r/a_0}\, dr.$$

$$= \frac{4}{a_0^3} \cdot \frac{4!}{a_0^5\, 2^5} \cdot a_0^5 = a_0^2 \cdot \frac{\cancel{4} \times 2\cancel{4}\, 3}{\cancel{32}\, \cancel{8}}$$

$$\boxed{\langle r^2 \rangle = 3a_0^2} \quad \text{(Ans)}$$

$$\therefore \boxed{\frac{r_{mp}}{\langle r \rangle} = \frac{2}{3}} \quad \text{(Ans)}$$

(A) $$\langle p_r \rangle = \int_0^\infty R^* p_r(R)\, r^2\, dr$$

$$= \frac{-i\hbar 4}{a_0^3} \int_0^\infty e^{-2r/a_0} \left(\frac{\partial}{\partial r} + \frac{1}{r}\right)\left(e^{-r/a_0}\right) r^2\, dr$$

$$= -\frac{4i\hbar}{a_0^3} \int_0^\infty e^{-r/a_0} \left[-\frac{1}{a_0} e^{-r/a_0} + \frac{e^{-r/a_0}}{r}\right] r^2\, dr$$

$$= -\frac{4i\hbar}{a_0^3} \int_0^\infty e^{-2r/a_0} \left(\frac{1}{r} - \frac{1}{a_0}\right) r^2\, dr$$

$$= -\frac{4i\hbar}{a_0^3} \int_0^\infty r\, e^{-2r/a_0} + \frac{4i\hbar}{a_0^4} \int r^2 e^{-2r/a_0}\, dr.$$

$$= -\frac{\cancel{4}i\hbar}{a_0^3} \times \frac{a_0^2}{\cancel{2}} + \frac{\cancel{4}i\hbar}{a_0^4} \cdot \frac{\cancel{2} \cdot \cancel{8}\, a_0^3}{\cancel{8}\, a_0} = 0. \quad \boxed{\langle p_r \rangle = 0}$$

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$$\therefore\ p_r^2 f = -\hbar^2 \left(\frac{\partial}{\partial r} + \frac{1}{r}\right)\left(\frac{\partial f}{\partial r} + \frac{f}{r}\right)$$

$$= -\hbar^2 \left[\frac{\partial^2 f}{\partial r^2} + \cancel{-\frac{f}{r^2}} + \frac{1}{r}\cdot\frac{\partial f}{\partial r} + \cancel{\frac{f}{r^2}} + \frac{\partial f}{\partial r}\cdot\frac{1}{r}\right] \quad \cancel{= -\hbar^2\left(\frac{\partial^2}{\partial r^2} + \right.}$$

$$= -\hbar^2\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\right) f$$

$$\Rightarrow \boxed{p_r^2 = -\hbar^2\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\right)}$$

$$\therefore\ \langle p_r^2 \rangle = \int_0^\infty R^* (p_r^2 R)\, r^2\, dr$$

$$= -\frac{4\hbar^2}{a_0^3}\int_0^\infty e^{-r/a_0}\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\right) e^{-r/a_0}\cdot r^2\, dr$$

$$= -\frac{4\hbar^2}{a_0^3}\int e^{-r/a_0}\left[-\frac{1}{a_0}\frac{\partial}{\partial r}\left(e^{-r/a_0}\right) - \frac{2}{r a_0} e^{-r/a_0}\right] r^2\, dr$$

$$= -\frac{4\hbar^2}{a_0^3}\int e^{-r/a_0}\left[+\frac{1}{a_0^2} e^{-r/a_0} - \frac{2}{a_0 r} e^{-r/a_0}\right] r^2\, dr$$

$$= -\frac{4\hbar^2}{a_0^3}\int_0^\infty e^{-2r/a_0}\left(\frac{1}{a_0^2} - \frac{2}{r a_0}\right) r^2\, dr$$

$$= -\frac{4\hbar^2}{a_0^5}\int_0^\infty e^{-2r/a_0} r^2 + \frac{8\hbar^2}{a_0^4}\int_0^\infty r\, e^{-2r/a_0}\, dr$$

$$= -\frac{\cancel{4}\hbar^2}{a_0^5}\cdot\frac{\cancel{2}}{\cancel{a_0}}\,\frac{a_0^3}{\cancel{8}} + \frac{\overset{2}{\cancel{8}}\hbar^2}{a_0^4}\cdot\frac{a_0^2}{\cancel{4}}$$

$$\boxed{\langle p_r^2 \rangle = \frac{\hbar^2}{a_0^2}}$$

$$\therefore\ \Delta p_r = \sqrt{\langle p_r^2 \rangle - \langle p_r \rangle^2} = \boxed{\frac{\hbar}{a_0} = \Delta p_r} \quad \text{(Ans)}$$